

Pay for Training, Not for Openness: Reward Design for Original and Distilled AI Models

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PS2 · CAPABILITY-TRANSFER OPENNESS

KEY MESSAGE Private and social rankings are exact reverses: the developer's best move is society's worst. So markets rest on secrecy, or fill with open copies, unless the prize is paid per unit of **training compute**, not for **openness** alone.

01 Question & model

Research question & verified gap

Q A developer may imitate (distil) or innovate (train from scratch), and may release the result under secrecy or openness. Which of the four does it choose — and does society want the same?

Gap Rules that reward openness reward it as such (Qiu et al. 2025), and stay silent on whether a model was originally trained or merely imitated. That silence is where the incentive breaks.

Why it matters A prize that pays for openness alone pays the imitator, not the innovator.

Strategic model & foundations

Players one developer chooses sourcing (imitate or innovate) and release (secrecy or openness).

Payoffs four cells (Figure 1): secrecy keeps the monopoly profit; a release earns a prize on its compute and loses the edge.

Solution concept Nash equilibrium, set against society's ranking.

Assumptions innovation costs more and uses more compute than imitation; the prize rate k is paid per unit of compute E .

Notation C_d , E_d = cost and compute of imitation · C_n , E_n = cost and compute of innovation · k = prize per unit of E · L = loss from disclosure · π = monopoly profit kept under secrecy.

Three complementary lenses

Game theory left alone, the best response is secrecy plus imitation — the cheap way to keep the monopoly price.

Social choice society ranks openness plus innovation best and secrecy plus imitation worst; the two orders are exact reverses.

Mechanism design a prize paid for openness alone rewards the imitator more than the innovator, who paid more to get there; paid per unit of compute it separates them.

Auction or allocation comparison

The prize is an **all-pay auction** (Baye et al. 1996): E is the bid, all bidders pay, the top E wins.

Rule A — flat prize an openness-only prize pays both open cells the same, so innovation never wins.

Rule B — prize per unit of compute two conditions: a firm discloses only if $kE_d > L + \pi - \beta\phi K$, and innovates only if $k(E_n - E_d) > C_n - C_d$.

Where the all-pay analogy bites the bid is real compute, verifiable only by audit, so over-reporting pays.

Future share with a strong motive firms open with no prize, but they open copies (fresh run: seed 206):

reward rule	(a) no motive, $\beta\phi K = 0$	(b) motive on, $\beta\phi K = 19.98$
none, $k = 0$	.96/00/04/00 ①	.28/68/01/03 ②
openness, $R = 20$	.11/85/00/04 ②	.01/95/00/04 ②
indexed, $k = 4$	.35/00/01/64 ③	.03/02/00/95 ③
indexed, capped, $k = 12$	.42/54/02/03 ②	.03/93/00/05 ②

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02 Design & evidence

Figure 1 — the four strategy combinations			
		Imitate · cheap (C_d , E_d)	Innovate · costly (C_n , E_n)
Secrecy	① Secrecy + imitation Free-rider — takes, gives nothing $\pi - C_d$		③ Secrecy + innovation Monopolist — builds, keeps it $\pi - C_n$
	② Openness + imitation Relay — takes and gives back $-C_d + kE_d - L + \beta\phi K$		④ Openness + innovation Source — builds, and shares $-C_n + kE_n - L + \beta\phi K$

Figure 1. The strategy space is a 2×2 — sourcing (imitate or innovate) $\times$ release (secrecy or openness) — with each cell's payoff. Secrecy plus imitation keeps the monopoly price cheaply, which is why it dominates privately; openness plus innovation is the only cell that creates a reusable source, and it is the most expensive. Cells recomputed in `verify_k.py`. An open release also keeps the future market share it buys, $\beta\phi K$ — the discounted value of the external contributions an open model attracts.

Figure 2 — private desire vs. social desirability			
Combination	Private desire	Social desirability	Conflict
① Secrecy + imitation	████████	███	← Largest conflict
③ Secrecy + innovation	██████	███	← Conflict
② Openness + imitation	██████	██████	← Eased
④ Openness + innovation	██████	████████	← Largest conflict

Dark red = private desire · green = social desirability (more squares = more wanted).

Three market states — the prize rate k decides which cell wins

Market state	Prize rate	Composition of the market	Outcome
State ① all secrecy	too small for disclosure	① dominates if $\beta\phi K$ is small	monopoly price; no source
State ② all imitation	disclosure, not innovation	② dominates — nobody innovates	no source is ever built
State ③ healthy	pays for both	④ wins; some secrecy survives	fair price and a live source

Figure 2. The private order (①>③>②>④) and the social order (④>②>③>①) are exact mirror images: the developer's favourite is society's least wanted. The private order holds while $\pi + L > C_n - C_d$ — what secrecy keeps, plus what disclosure loses, must outweigh the extra cost of innovating.

Interpretation. A prize too small buys no disclosure; one that pays for openness but not for cost buys copies, not capability; only a prize covering both keeps a live source.

Verification. Cells and states are recomputed from scratch in the checker, independently of the symbolic derivation; an interactive page lets a reader move the parameters and watch the winning cell flip.

Runs. Paying for openness alone fills the market with imitation and dries up the sources; paying per unit of compute turns it back to original release (sources: 0.04 $\rightarrow$ 0.95); capping the prize undoes that, and with the future-share motive opening pays only for $m \in (0.101, 1.519)$. Fresh run: seed 206. Copying a copy adds no capability, so mutual imitation burns compute while capability decays — yet some imitation is still efficient, because it stops every firm from re-deriving the same result. Parameters are illustrative, not estimated.

Selected references

[1] Habibi, M. (2025). Open Sourcing GPTs: Economics of Open Sourcing Advanced AI Models. arXiv:2501.11581.
[2] Xu, F., Wang, X., Chen, W. & Xie, K. (2025). The Economics of AI Foundation Models: Openness, Competition, and Governance. arXiv:2510.15200.
[3] Nalebuff, B. & Stiglitz, J. (1983). Prizes and Incentives: Towards a General Theory of Compensation and Competition. Bell Journal of Economics 14(1), 21–43.
[4] Baye, M. R., Kovenock, D. & de Vries, C. G. (1996). The All-Pay Auction with Complete Information. Economic Theory 8(2), 291–305.
[5] Qiu, T., Laufer, B., Kleinberg, J. & Heidari, H. (2025). Modeling the Economic Impacts of AI Openness Regulation. NeurIPS 2025.

03 Impact & dialogue

Practical impact: a real-world case

Case an agency that pays for open-weight models, such as a national compute subsidy, without asking whether the model was trained or copied.

Decision informed index the payment to training compute, not to openness: openness alone is cheap to satisfy by imitating.

Who gains, who bears risk innovators gain; imitators lose a payment they never earned; users gain a live source instead of a copy of a copy.

Feasibility training compute is auditable from training records.

Challenge route compute is a contested proxy for capability; the prize rate is a political choice.

Interdisciplinary evidence

Economics — mechanism design the rank reversal shows the market cannot self-select openness plus innovation, so a per-unit prize restores the incentive (Nalebuff & Stiglitz 1983).

Computer science — machine learning imitation is a standard, low-cost technique, so openness is cheap to fake, while training compute is measurable and can serve as the index. Copying is cheap but never free, so imitation still belongs in an efficient mix.

Boundary on the claim one representative developer; compute treated as verifiable; no multi-firm race.

Integration cheap copying breaks the economics; measurable compute repairs it.

Artifacts the symbolic derivation, a zero-dependency recheck, an interactive game, the repository.

Limitations & next test

Main limitation single-firm and static: it shows where the market settles, not how fast.

Next test add a rival and a cost of detecting imitation, and check whether open innovation survives when training compute must be measured imperfectly.

Question for reviewers if reported compute can be inflated, does the per-unit prize collapse back into all-imitation?

SEP 28 Symposium, peer evaluation & case impact

SEP 30 Final update

OPEN MATERIALS

Hugging Face: huggingface.co/spaces/dku-comsci-econ206-2026/FP4_PS2
GitHub: github.com/dku-comsci-econ206-Autumn2026/PS2-FP4-CapabilityTransfer
Commit 【填: 最终 commit】 · MIT licence · scan to run the Prize Rate Game

